

Consumer Valuation of Battery Health Information in Used Electric Vehicle Markets: A Discrete Choice Experiment

Xiatian Iogansen^{a,b,1}, Zain Hoda^{b,2}, Christina Gore^{a,*,3}, Joshua D. Kneifel^{a,4}, Sindhu Ranganath^{a,b,5} and John P. Helveston^{b,6}

^aApplied Economics Office, Engineering Laboratory, National Institute of Standards and Technology, Maryland, USA

^bDepartment of Engineering Management and Systems Engineering, George Washington University, Washington D.C., USA

ARTICLE INFO

Keywords:

Battery electric vehicle; Used vehicle market; Discrete choice experiment; Willingness-to-pay; Consumer preferences; Information treatment

ABSTRACT

The used battery electric vehicle (BEV) market is growing, yet consumers face substantial uncertainty about battery condition at the point of sale. This paper estimates consumer preferences and willingness to pay (WTP) for vehicle attributes in the used BEV market using a discrete choice experiment (DCE) administered to a national sample of 3,072 U.S. adults. The DCE jointly varies conventional vehicle attributes (price, mileage) and battery-specific attributes (electric range, degradation rate, refurbishment history) across six repeated choice tasks. A mixed logit model confirms substantial preference heterogeneity at the population level, with consumers valuing vehicles with lower mileage and additional range, while penalizing range degradation and battery refurbishment. The six-class latent class choice model reveals six distinct consumer segments that differ markedly in their sensitivity to battery quality signals, price, and the no-choice option, which can be further explained by their distinctive characteristics. An information treatment analysis further shows that providing detailed battery replacement cost and warranty information does not affect overall market (dis)engagement, but reshapes the type of concern driving opt-out behavior. These findings carry direct implications for used vehicle dealers, OEMs, and certification service providers seeking to develop market infrastructure for used BEVs.

1. Introduction

The used vehicle market plays a central role in U.S. automobile ownership: used vehicle transactions now account for roughly 70% of all passenger vehicle purchases and are projected to grow steadily through the coming decade (Market Research Future, 2025). As electric vehicles (EVs) continue to penetrate the new car market, an increasing number of BEVs are entering the secondary market. Used EV sales rose 34.2% year-over-year through early 2025, while average days of supply fell by 21.5%, signaling rapid and accelerating demand (Cox Automotive Inc., 2025). The growth of the used BEV market extends EV access to middle- and lower-income households (Hagman et al., 2016), has the potential to support a circular economy by extending vehicle and battery lifetimes (Guzek et al., 2024), and underpins the residual values that make new EV leases financially viable (Brückmann et al., 2021; Lim et al., 2015).

Despite this momentum, consumer hesitation toward used BEVs remains pronounced. Fernandes (2023) indicates that 52% of U.S. adults are unlikely to consider purchasing a used EV, compared to only 28% who express openness to the idea. Consumers frequently cite *uncertainty about vehicle and battery condition* (41%) and *concerns about reliability relative to new EVs* (23%) as primary barriers. Among those who have already purchased a used EV, over half reported pre-purchase concerns about battery longevity, even though only 5.1% experienced substantial performance loss in practice (Sheykhfard et al., 2025). This gap between perceived and actual risk points to a fundamental information asymmetry: buyers lack information and knowledge about battery health at the point of sale, even though the battery can represent up to 30% of a used BEV's value (Boudway, 2020).

*Corresponding author

¹xiatian.iogansen@gwu.edu & xiatian.iogansen@nist.gov, <https://orcid.org/0000-0002-4851-1323>

²zain.hoda@gwu.edu

³christina.gore@nist.gov, <https://orcid.org/0000-0002-3586-6918>

⁴joshua.kneifel@nist.gov

⁵sindhu.ranganath@nist.gov, <https://orcid.org/0000-0001-5764-9773>

⁶jph@gwu.edu, <https://orcid.org/0000-0002-2657-9191>

1 Unlike conventional vehicles, where odometer reading and vehicle age serve as reasonable proxies for overall wear,
2 battery degradation in BEVs depends on factors such as charging history, depth of discharge cycles, thermal exposure,
3 and driving patterns (Bashash et al., 2011; Neubauer et al., 2012). Two BEVs of identical age and mileage can differ
4 substantially in their remaining battery capacity and projected useful life. The problem is that this information is
5 less visible to buyers, and some of it is not even available to sellers themselves. As a result, sellers cannot credibly
6 signal battery quality and struggle to recoup the true value of their vehicles at resale, while buyers cannot make
7 informed decisions, often undervaluing well-maintained EVs or unknowingly overpaying for poorly maintained ones.
8 The consequences extend beyond individual transactions: persistent market uncertainty may suppress new EV adoption
9 by generating "resale anxiety" among prospective buyers who question whether their new vehicle will hold its value
10 (Brückmann et al., 2021).

11 On the information supply side, dealer- and consumer-facing battery-health reports are commercially available
12 via battery-health analytics firms, with an option for third-party certification (Aviloo 2026; Battery-Tech Network,
13 2025; Recharged, 2026; Recurrent Auto, 2025). However, these reports alone are not enough to resolve used-BEV
14 information asymmetry, for several reasons: (1) much battery management system (BMS) data is gated by original
15 equipment manufacturers (OEMs) and coverage differs across OEMs, and no firm has a path to comprehensive cross-
16 make BMS access without regulatory change; (2) different firms operate different certification regimes with non-
17 comparable scoring methodologies; (3) certification is non-trivial and the cost may be prohibitive for many used-BEV
18 sellers, especially those with lower-value vehicles, where the information asymmetry is most damaging. An equally
19 important question is whether consumers value battery health information enough to pay for it, and if so, what type of
20 information they value and by how much. Answering this question may help justify the need to standardize information
21 disclosure for used BEVs in the long term, and is the focus of this paper.

22 The study explores vehicle buyers' perspectives by administering a discrete choice experiment (DCE) to a national
23 sample of 3,072 U.S. adults who reported an intention to purchase a used car or SUV within the next two years. The
24 DCE presents respondents with repeated choices among hypothetical used BEVs that vary in mileage, purchase price,
25 electric range, battery degradation rate, and battery refurbishment history. Respondents are also randomly assigned
26 to receive either basic range-and-health information or extended information that additionally describes battery
27 maintenance costs and warranty coverage, allowing us to assess whether information treatment shapes preferences and
28 opt-out behavior. We estimate two complementary models. A mixed logit (MXL) model in WTP space characterizes
29 aggregate preferences and the degree of preference heterogeneity across the full sample. A six-class latent class choice
30 model (LCCM) then identifies discrete consumer segments that differ in their sensitivity to vehicle attributes. To our
31 knowledge, this study provides the first WTP estimates for battery health information in the U.S. used BEV market.
32 We also implemented a thematic analysis of open-ended responses from respondents who opted out of all six choice
33 tasks, providing additional insight into the reasons for market disengagement.

34 The MXL results confirm that the population, on average, values additional electric range (\$11,070 per 100 miles
35 at Year 3), penalizes higher mileage (\$2,780 per 10,000 miles), and discounts battery degradation rate (approximately
36 \$980 per percentage point of annual range loss). Both refurbishment types carry significant disutility, though large
37 standard deviation estimates indicate substantial variation across individuals. The LCCM reveals that this heterogeneity
38 is structurally organized: approximately 62% of respondents belong to three classes that strongly penalize any
39 refurbishment history. A further 17% are either price-constrained buyers largely indifferent to battery quality signals
40 or deeply skeptical non-adopters unlikely to enter the used BEV market in the near term. The remaining class consists
41 of highly committed, price-insensitive buyers who place an exceptionally high premium on a pristine battery service
42 history. The information treatment analysis reveals that extended battery information raises the overall disinterest in
43 used BEVs and the salience of battery-specific concerns among opt-outs, but does not reduce the overall rate of market
44 disengagement.

45 These findings are relevant to dealers, OEMs, battery service providers, and financing institutions that are
46 developing commercial infrastructure for used BEVs. The existence of large, identifiable segments with strong
47 preferences for battery quality documentation supports the viability of premium pricing for certified battery health,
48 while the presence of a price-sensitive but engaged segment points to the importance of affordable product offerings.

49 The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes
50 the survey design and sample. Section 4 presents the econometric framework. Section 5 reports results and discusses
51 market and business implications. Section 6 concludes.

2. Literature Review

This section reviews key literature informing survey design and analyses of this study. The first subsection examines the main drivers and barriers to the adoption of used BEVs, while the second subsection focuses on EV battery technology and how it shapes consumer perceptions and decision-making.

2.1. Used Electric Vehicle Adoption

Gore et al. (2024) provided a comprehensive review on consumer perspectives on EVs and the factors influencing EV adoption, identifying a broad range of determinants including attitudinal, demographic, societal, environmental, technological, and economic variables (Iogansen et al., 2023; Naseri et al., 2024; Sonar et al., 2023). However, most existing studies focused on new EV markets, with comparatively limited attention given to the used EV markets, which is rapidly expanding and playing an increasingly important role in the broader diffusion of electric mobility.

Though some preferences may carry over between new and used vehicle buyers—for instance, longer driving range consistently correlates with stronger EV preference—emerging evidence suggests that these two consumer groups differ in meaningful ways. For instance, used EV buyers are more sensitive to the availability of public charging infrastructure than new EV buyers, likely due to lower rates of home charging access (Zou et al., 2020). Moreover, purchasing behavior for used vehicles is often more risk-averse and budget-conscious, reflecting different decision-making processes than those found in the new car market. This indicates that used EV adoption is not simply a scaled-down version of new EV adoption and warrants independent investigation.

Several studies have begun to profile the typical used EV buyer. Canepa et al. (2019) found that used car buyers are commonly cost-conscious. Compared to the general population, used EV buyers are more likely to be younger, male, White, and better educated. However, compared to new-EV buyers, used EV buyers tend to have lower household incomes and be renters, with lower access to private garages with EV chargers (Khaloei et al., 2020; Sheykhfard et al., 2025). These characteristics differ from the traditional early adopter profile seen in the new EV market (typically wealthier homeowners with garages), suggesting that used EVs may serve as a more accessible entry point into EV ownership for broader population segments.

Although range is a common concern for EV users, Sheykhfard et al. (2025) found that the majority of used EV owners drive well within the range capabilities of most existing EV models: 59.4% reported driving fewer than 100 miles per day and 29.4% drove less than 50 miles per day. This implies that range anxiety may be less of a practical barrier for used EV buyers, who tend to have shorter commutes or use their vehicles in more limited contexts. Range may therefore be a less decisive factor in used EV adoption compared to other attributes such as battery condition, charging accessibility, and price.

Electric vehicle batteries (EVBs) present new risks and opportunities for both consumers and the automotive industry. Despite progress in battery technology, consumer concerns about battery health, replacement cost, and resale value persist, particularly in the used EV market. Battery costs can represent up to 30% of a vehicle's total price (Boudway, 2020), making battery longevity and reliability central to the economic attractiveness of used EV ownership.

2.2. Battery Degradation and State-of-Health

In the EV adoption literature, battery-related considerations have largely been operationalized through driving range (Yuan et al., 2018). Prospective used EV buyers commonly express concerns about battery degradation, SOH uncertainty, and the perceived unreliability of older battery technology (Pedrosa and Nobre, 2018). Battery degradation manifests as a reduction in energy capacity, driving range, and charging efficiency over time (Attia et al., 2022). The rate of degradation depends on charging patterns, driving behavior, depth of discharge, and operating conditions such as temperature exposure (Bashash et al., 2011; Neubauer et al., 2012). Yang et al. (2018) estimates that BEV batteries can last between five and thirteen years depending on climate conditions. Battery aging trajectories tend to be linear or sublinear, as degradation rates often stabilize after initial use due to the settling of battery chemistry (Attia et al., 2022, 2020). A battery is generally considered to have reached its End-of-Life (EoL) for vehicle use when its SOH drops to approximately 70–80% of original capacity (Canals Casals et al., 2022, 2019). Battery degradation reduces the perceived future value of a used BEV and heightens resale anxiety among prospective buyers.

2.3. Battery Replacement, Cost, and Warranty Considerations

While the literature has examined battery longevity in the context of total cost of EV ownership (Hagman et al., 2016; Letmathe and Soares, 2017), direct evidence on the cost of battery replacement in used BEVs remains limited. Most modern EVBs are designed to outlast the vehicle's useful life under normal use. Dnistran (2024), in a study

of nearly 5,000 EVs, estimated an average annual degradation rate of just 1.8%. Nevertheless, when replacement is required, costs can range from \$5,000 to \$20,000 or more depending on vehicle model and labor (Kothari, 2024; Witt, 2024). To support consumer confidence, most EV manufacturers provide warranties covering 8 years or 100,000 miles, whichever comes first (Clarke, 2024). For used vehicle buyers evaluating whether a battery's SOH will remain above the 70–75% warranty threshold within their ownership window, the presence of remaining warranty coverage represents meaningful additional value. Letmathe and Soares (2017) further account for battery residual value as a separate component of total cost of ownership, underscoring the economic significance of battery condition at resale.

2.4. Second-Life and Refurbished Batteries

As the number of retired EVBs increases, second-life applications and refurbishment are gaining commercial attention (Skeete et al., 2020; Tankou et al., 2023). The two dominant second-life pathways are repurposing degraded batteries for stationary energy storage and refurbishing them for continued use in EVs. This paper focuses on the latter. Refurbishing batteries at the cell or module level can lower vehicle costs and support the commercial viability of BEVs: Jiao and Evans (2016) show that repurposing EoL EVBs could reduce the cost gap between BEVs and conventional vehicles, while Shaikh et al. (2023) demonstrate through simulation that refurbished EVBs are less expensive than new ones, with cost advantages that are greatest in local markets. However, existing research on second-life EVBs has focused primarily on technical and economic feasibility, with comparatively little attention to consumer demand for refurbished options in the U.S. context.

One of the few consumer-facing studies, conducted in Portugal, found that respondents expressed positive attitudes toward used EVs fitted with replacement batteries backed by dealership warranties, and indicated higher WTP relative to vehicles with original batteries (Pedrosa and Nobre, 2018). This qualitative study was small in scope and may not generalize to the U.S. market. More importantly, no study to date has examined consumer valuations of partial versus full battery replacement, or estimated WTP for battery health disclosure as distinct from the battery condition itself. These gaps motivate the present study.

2.5. Willingness to Pay for Battery Information

To be continued.

3. Survey Design, Data Collection, and Sample Profile

3.1. Nationwide Used EV Survey

The survey was designed to elicit consumer preferences and WTP for vehicle attributes and battery health information in the used BEV market. The target population consists of U.S. adults aged 18 or older who reported an intention to purchase a used car or SUV within the next two years. For a full description of the survey instrument, see Gore et al. (2026).⁷

The centerpiece of the survey is a BEV discrete choice experiment with six repeated choice tasks in which respondents choose among hypothetical used BEVs with varying attributes, with particular attention to how battery health information is valued. All vehicles are assumed to have been manufactured in 2022 and to be three years old at the time of purchase. The varying attributes are vehicle mileage, purchase price, electric range and battery SOH, and battery refurbishment history. Prior to the choice tasks, respondents are shown descriptions of all attributes they will encounter, including detailed definitions for battery-specific terms (see Table 1). During the DCE design phase, we used range at the time of manufacture (Year 0) and an annual degradation rate to characterize battery condition. To improve comprehension for respondents, we transformed these underlying design parameters and presented range and SOH at Year 3 (at the point of purchase) and Year 8 (five years after purchase).

To examine how battery information treatment shapes preferences, respondents were randomly assigned to one of two treatment conditions. The *basic information* group received descriptions of electric range and battery SOH only. The *extended information* group additionally received information about expected battery useful lifetime, replacement cost, and warranty coverage. This design allows us to test whether the content of battery information affects consumers' choices and their propensity to opt out. Figure 1 shows an example choice task, in which respondents select among three used BEVs or a "None of the above" option.

⁷The Data Collection Instrument was published prior to data collection. Certain aspects of the survey design were revised following piloting. Where discrepancies exist between the instrument and this paper, the description in this paper reflects the final implemented version.

Table 1
BEV choice experiment attributes and levels.

Attribute	Definition	Levels			
		Low-budget (≤\$20K) Car	High-budget (>\$20K) Car	Low-budget (≤\$20K) SUV	High-budget (>\$20K) SUV
Mileage	The total number of miles a vehicle has traveled since manufacture.	15,000–60,000 miles, in increments of 500 miles.			
Battery refurbishment history	The repair or replacement work that has been performed on the vehicle's main battery pack.	(1) <i>Original</i>: The battery remains in its factory-original condition with no repairs or replacements. (2) <i>Some battery cells replaced</i>: A portion of the battery cells has been replaced to address performance issues or extend battery life; the remaining cells are original. (3) <i>Entire battery pack replaced</i>: The full battery pack has been replaced with a new, refurbished, or used pack, typically after battery performance declined below a specified threshold.			
Electric range at delivery (Year 0)	The maximum distance the vehicle can travel on a full battery charge under typical driving conditions, as rated at the time of manufacture.	50–150 miles (50-mile increments)	100–250 miles (50-mile increments)	150–250 miles (50-mile increments)	200–350 miles (50-mile increments)
Battery annual degradation rate and state of health (SOH)	The average annual percentage loss in electric range and battery SOH. The remaining range and SOH at Year 3 (current) and Year 8 (five years from purchase) are derived from this rate.	1%–8%, in increments of 1%.			
Purchase price	The total cost of the vehicle in dollars, including down payment, monthly payments, taxes, and fees.	\$10,000–\$20,000 (\$2,000 increments)	\$20,000–\$40,000 (\$5,000 increments)	\$15,000–\$25,000 (\$2,000 increments)	\$25,000–\$45,000 (\$5,000 increments)

Assume you are able to purchase a 3-year-old used vehicle that looks like the one you selected (see photo). Which of the following versions of that vehicle would you be most likely to purchase? Please compare the key features shown for each option before making your selection.



Figure 1: Example of a choice experiment.

In addition to the DCE, the survey collects information across three broader topic areas. Before the choice tasks, respondents report on their current household vehicle fleet (including fuel type, range, acquisition method, and costs) and their future vehicle purchasing plans (including anticipated budget, payment method, and the likelihood of purchasing a new or used plug-in hybrid or BEV). After the choice tasks, the survey assesses respondents' knowledge, attitudes, and perceptions regarding EVs and battery technology. This section includes factual questions about EV charging capabilities and available federal tax incentives, as well as psychological statements rated on five-point Likert scales. The psychological items cover social norms, environmental beliefs, cost perceptions, battery concerns, and personality traits such as risk-taking and price sensitivity. Two items about refurbished EV batteries are of particular relevance: "Purchasing refurbished EV batteries will minimize negative effects on natural ecosystems" (hereafter "environmentally positive") and "Refurbished EV batteries do NOT perform and function as original EV batteries" (hereafter "functionally negative"). The final survey section collects demographic information including age, gender, race and ethnicity, household size, income, education, employment status, and housing type.

3.2. Data Collection and Sample Profile

The survey was administered online through two widely-used research platforms (Qualtrics and Dynata) between December 2025 and April 2026. The median completion time was approximately 14 minutes. All participants received platform-standard compensation.

The initial sample comprised 3,657 respondents from across the U.S. We applied multiple data quality screens to identify and remove low-quality responses, including speeders, respondents who failed the embedded attention check, those with internal inconsistencies across survey sections, and those who provided uninformative or incoherent text in open-ended responses. For the DCE in particular, we applied an additional speeder screen designed to identify inattentive responders.

The final sample consists of 3,072 respondents, yielding 18,432 choice observations across six tasks. Figure 2 shows the geographic distribution of respondents by ZIP code; point size and color intensity are scaled to respondent count, indicating broad national coverage.

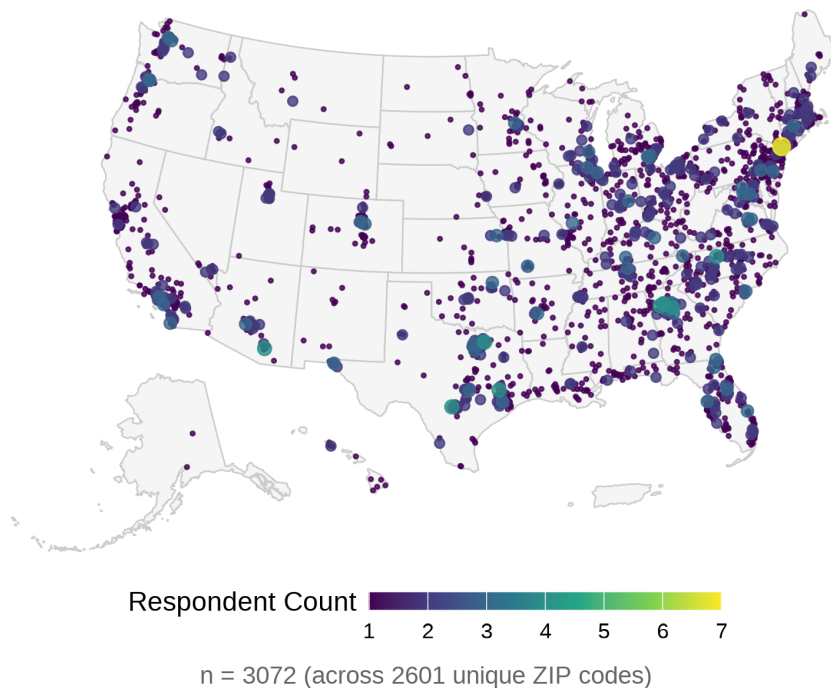


Figure 2: Geographic distribution of survey respondents across the continental United States.

4. Method

4.1. Mixed Logit Model

We begin by estimating WTP for changes in vehicle attributes across the full sample (3,072 respondents, 18,432 choice observations). This provides an aggregate baseline and reveals the extent of preference heterogeneity. We estimate a Mixed Logit (MXL) model directly in *WTP space* using the *logitr* R package (Helveston, 2023). The MXL model relaxes the Independence of Irrelevant Alternatives (IIA) assumption of the standard Multinomial Logit (MNL) model by allowing preference parameters to vary continuously across individuals (McFadden and Train, 2000; Train, 2009). Estimating in WTP space allows us to directly specify the distribution of WTP for each non-price attribute, rather than recovering it post hoc from the ratio of attribute to price coefficients. We assume that individual WTP values follow normal distributions, capturing symmetric heterogeneity around population means. The utility parameterization follows the notation in Helveston et al. (2018) and Helveston (2023).

The WTP-space utility for respondent n choosing alternative i in choice scenario t is:

$$u_{nti} = \lambda_n (\omega_n^\top \mathbf{x}_{nti} - p_{nti}) + \varepsilon_{nti}, \quad \varepsilon_{nti} \sim \text{Gumbel}\left(0, \frac{\pi^2}{6}\right) \quad (1)$$

where p_{nti} is the purchase price presented in scenario t to respondent n ; $\mathbf{x}_{nti}$ is a vector of non-price BEV attributes including vehicle mileage, battery refurbishment status, current electric range at the point of purchase (Year 3), and the proportional rate of range loss over the subsequent five years (to Year 8);⁸ $\lambda_n > 0$ is the individual-specific scale parameter; ω_n is a vector of individual WTP values drawn from a mixing distribution $f(\omega \mid \mu, \Sigma)$ with population mean μ and covariance Σ ; and ε_{nti} is an IID Type I Extreme Value error. Because λ_n and ω_n are both unobserved, the choice probability requires integrating the logit kernel over their joint mixing distribution:

$$P_{nti}(\mu, \Sigma) = \int \int \frac{\exp(\lambda(\omega^\top \mathbf{x}_{nti} - p_{nti}))}{\sum_j \exp(\lambda(\omega^\top \mathbf{x}_{ntj} - p_{ntj}))} f(\omega \mid \mu, \Sigma) g(\lambda) d\omega d\lambda \quad (2)$$

This integral is evaluated by simulation using Halton draws ($R = 500$). Accounting for the panel structure of six repeated choice tasks per respondent, the simulated log-likelihood is:

$$\mathcal{L}(\mu, \Sigma) = \sum_{n=1}^N \ln \left[\frac{1}{R} \sum_{r=1}^R \prod_{t=1}^T \frac{\exp(\lambda_r(\omega_r^\top \mathbf{x}_{nti_t^*} - p_{nti_t^*}))}{\sum_j \exp(\lambda_r(\omega_r^\top \mathbf{x}_{ntj} - p_{ntj}))} \right] \quad (3)$$

where (λ_r, ω_r) is the r -th draw from the joint mixing distribution and i_t^* is the alternative chosen by respondent n in scenario t . Because estimation is carried out directly in WTP space, the population mean WTP for each non-price attribute k is obtained directly as $\hat{\mu}_k$, without requiring a post-estimation ratio transformation.

4.2. Latent Class Choice Model

Model Framework

The MXL results reveal substantial preference heterogeneity across respondents. To explore whether this heterogeneity reflects discrete consumer segments with systematically different preferences, we estimate a Latent Class Choice Model (LCCM) using the *Apollo* R package. The LCCM assumes that individuals can be assigned to a finite number of unobserved groups (classes) within which preferences are homogeneous, but across which they differ (Greene and Hensher, 2003; Kamakura and Russell, 1989). This structure is well suited to our research context, where consumers may differ fundamentally in their attitudes toward battery health risk, their familiarity with EV technology,

⁸We compared six attribute specifications that varied whether range was expressed at Year 0 or Year 3, and whether battery health change was expressed as absolute range loss or as a percentage degradation rate. The specification combining range at Year 3 with percentage loss rate from Year 3 to Year 8 yielded the best model fit and the most economically interpretable coefficients, suggesting consumers evaluate used BEVs primarily in terms of current performance and relative future deterioration.

and their price sensitivity. The model is estimated on a subsample obtained after excluding cases with missing data on covariates (2,691 respondents, 17,496 choice observations).

As illustrated in Figure 3, the LCCM consists of two simultaneously estimated components. The *class membership model* assigns each respondent probabilistically to a latent class as a function of observable individual characteristics. The *class-specific choice model* then estimates the conditional probability of each vehicle choice within each class. In addition, a set of *inactive indicators*, including attitudinal and behavioral variables excluded from estimation due to potential collinearity or theoretical considerations, are used to further characterize the classes descriptively after estimation. The formal model structure follows the notation of Chen et al. (2023).

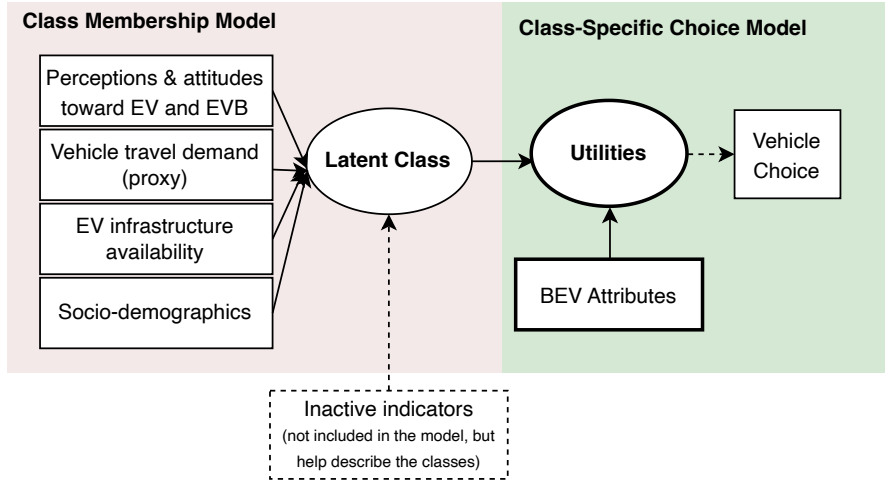


Figure 3: LCCM model framework.

Class-specific Choice Model

For respondent n belonging to latent class s ($s = 1, 2, \dots, S$), the utility of choosing vehicle alternative i in choice scenario t ($t = 1, \dots, 6$) is:

$$U_{nti|s} = \beta'_s \mathbf{X}_{nti} + \varepsilon_{nti|s} \quad (4)$$

where $\mathbf{X}_{nti}$ is a vector of observed BEV attributes (vehicle mileage, battery refurbishment status, electric range at Year 3, proportional range loss to Year 8, and purchase price); β_s is a vector of class-specific utility parameters; and $\varepsilon_{nti|s}$ is an IID Type I Extreme Value error. Under this assumption, the conditional probability that respondent n in class s chooses alternative i in scenario t follows the MNL form:

$$P_{nt}(i | s) = \frac{\exp(\beta'_s \mathbf{X}_{nti})}{\sum_{j \in C_{nt}} \exp(\beta'_s \mathbf{X}_{ntj})} \quad (5)$$

where C_{nt} denotes the choice set available to respondent n at scenario t .

To capture the non-linear relationship between range and choice probability, we adopt a piecewise linear specification for the range and degradation rate attributes. Consumers likely have diminishing sensitivity to range at higher levels and heightened sensitivity around critical viability thresholds. We compared a quadratic specification, which imposes a symmetric parabolic shape and performs poorly at the tails, with a piecewise linear specification that estimates separate slopes for distinct segments. The piecewise specification provides superior model fit and more

interpretable coefficients. The breakpoints are defined as follows: electric range at Year 3 is segmented into 40–130 miles, 130–200 miles, and above 200 miles; the range loss rate is segmented into below 12%, 12–24%, and above 24%. These thresholds reflect both theoretical considerations about minimum driving viability and empirical patterns in the data.

Class Membership Model

The class membership model specifies the probability that respondent n belongs to latent class s as a function of observed individual characteristics:

$$M_n(s) = \frac{\exp(\gamma'_s \mathbf{Z}_n)}{\sum_{s'=1}^S \exp(\gamma'_{s'} \mathbf{Z}_n)} \quad (6)$$

where $\mathbf{Z}_n$ is a vector of individual-level covariates organized into four groups: (1) *perceptions and attitudes toward EVs and EV batteries* (e.g., range anxiety, environmental and functional assessments of refurbished batteries); (2) *vehicle travel demand proxies* (e.g., range of the respondent's primary vehicle); (3) *EV infrastructure availability* (e.g., access to electrical outlets for charging); and (4) *socio-demographic characteristics* (e.g., household income, vehicle type preference). The parameters for the reference class S are normalized to zero for identification.

The unconditional probability that respondent n chooses alternative i in scenario t , integrating over all latent classes, is:

$$P_{nt}(y_n = i) = \sum_{s=1}^S P_{nt}(i | s) \cdot M_n(s) \quad (7)$$

Substituting Equations (5) and (6) into Equation (7) gives the full integrated choice probability:

$$P_{nt}(y_n = i) = \sum_{s=1}^S \left[\frac{\exp(\beta'_s \mathbf{X}_{nti})}{\sum_{j \in C_{nt}} \exp(\beta'_s \mathbf{X}_{ntj})} \cdot \frac{\exp(\gamma'_s \mathbf{Z}_n)}{\sum_{s'=1}^S \exp(\gamma'_{s'} \mathbf{Z}_n)} \right] \quad (8)$$

The log-likelihood across all N respondents and T choice scenarios is:

$$\mathcal{L}(\beta, \gamma) = \sum_{n=1}^N \sum_{t=1}^T \ln P_{nt}(y_n = i_n^*) \quad (9)$$

where i_n^* denotes the alternative chosen by respondent n in scenario t . The model parameters β_s and γ_s are estimated jointly via maximum likelihood.

Willingness to Pay

A key output of the LCCM is class-specific WTP for each BEV attribute. Under the assumption that the marginal utility of price is negative and constant across alternatives, the WTP for attribute k in class s is:

$$\text{WTP}_{s,k} = - \frac{\hat{\beta}_{s,k}}{\hat{\beta}_{s,\text{price}}} \quad (10)$$

This represents the maximum dollar amount a representative respondent in class s would pay, or accept as compensation, for a one-unit change in attribute k , holding all other attributes constant.

4.3. Thematic Analysis of Reasons for Opt-out

For respondents who opted out systematically (i.e., did not choose any of the presented used BEV options in all six choice tasks; $n = 209$), our survey asked them to explain the reason(s) in an open-ended question at the end of the DCE. Complementary to the quantitative analyses discussed above, analyzing the wide range of open-text expression may help highlight the specific attributes that participants dislike most. It may also help us identify barriers that are not included in our attribute list or are not easily quantifiable via close-ended questions.

This study uses thematic analysis, which applies a coding system to organize qualitative information and identify patterns and themes (Braun and Clarke, 2006). By reviewing response texts, we constructed a two-level codebook of 11 parent themes and 25 sub-themes. Respondents could cite multiple reasons for opting out, suggesting potential interdependency among barriers (Krishna, 2021). We plot the theme network to visualize the co-occurrence of opt-out themes and identify clusters of related barriers. We further examine whether the battery information treatment affects respondents' opt-out rate and their reasons for opting out, by plotting the share of opt-out respondents who mentioned each theme and subtheme.

5. Results and Discussion

5.1. Results of Mixed Logit Model

Table 2 presents the MXL estimates. All mean WTP parameters are significant at the 0.1% level (***). On average, respondents value an additional 100 miles of electric range at Year 3 at \$11.1k, penalize every 10,000 miles of odometer reading by \$2.8k, and discount each additional percentage point of annual range-loss rate by approximately \$1.0k. Both refurbishment types generate meaningful disutility: pack replacement reduces WTP by \$4.1k and cell replacement by \$4.5k. The no-choice opt-out option carries a large negative coefficient (\$51.3k), reflecting a strong baseline preference to purchase a vehicle rather than opt out.

The standard deviation estimates confirm substantial preference heterogeneity across respondents. Range at Year 3 exhibits the widest dispersion in absolute terms ($SD = \$14.0k$ relative to a mean of \$11.1k), with the distribution of simulated WTP draws spanning from \$1.7k at the first quartile to \$20.5k at the third quartile. This indicates that while most respondents positively value range, a non-trivial share places little weight on it. Heterogeneity in the refurbishment parameters is also pronounced: the standard deviations for pack replacement (\$7.3k) and cell replacement (\$6.7k) are both larger in magnitude than their respective means, and the upper quartiles of the simulated distributions reach near zero (\$0.9k and \$0.1k, respectively). This implies that roughly one quarter of consumers view battery refurbishment neutrally or even as a positive quality signal, potentially because a replaced battery is perceived as newer than the original. Mileage dispersion is also substantial, with approximately one quarter of the simulated distribution crossing zero. In contrast, range-loss rate is the most uniformly penalized attribute: its standard deviation is small relative to the mean, and the distribution remains negative throughout nearly the entire sample. Together, these results confirm that consumer valuations of battery-related attributes are heterogeneous in both sign and magnitude, motivating the segment-level analysis that follows.

Table 2

Mixed Logit Model Estimates in WTP Space (price unit: \$10,000).

Parameter	Estimate	Std. Error	<i>p</i> -value				
<i>Mean parameters</i>	<i>Summary of 10k Draws</i>						
					1st Qu.	Median	3rd Qu.
λ (scale)	0.987	0.025	<0.001	***			
Mileage	-0.278	0.015	<0.001	***	-0.548	-0.277	-0.006
Range (Year 3)	1.107	0.045	<0.001	***	0.166	1.107	2.047
Range loss Rate (Year 3 to Year 8, in percentage)	-0.098	0.003	<0.001	***	-0.150	-0.098	-0.046
Battery Refurbishment: Pack Replace	-0.406	0.037	<0.001	***	-0.899	-0.407	0.085
Battery Refurbishment: Cell Replace	-0.448	0.037	<0.001	***	-0.899	-0.447	0.007
No choice (opt-out)	-5.130	0.110	<0.001	***			
<i>Standard deviation</i>							
SD: Mileage	-0.402	0.017	<0.001	***			
SD: Range (Year 3)	1.395	0.053	<0.001	***			
SD: Range loss Rate (Year 3 to Year 8, in percentage)	-0.077	0.003	<0.001	***			
SD: Battery Refurbishment: Pack Replace	0.730	0.060	<0.001	***			
SD: Battery Refurbishment: Cell Replace	-0.672	0.061	<0.001	***			
Observations					3,072 $\times$ 6 = 18,432		
Parameters					12		
Log-Likelihood					-18,615.44		
Null Log-Likelihood					-25,552.18		
AIC					37,254.89		
BIC					37,348.75		
McFadden R^2					0.2715		
Adj. McFadden R^2					0.2710		
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1.							

5.2. Results of Latent Class Choice Model**5.2.1. Model Selection**

To determine the optimal number of latent classes, we estimated models with one through eight classes (Table 3). Model fit was evaluated using log-likelihood (LL), the ρ^2 measure relative to the null model, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), adjusted ρ^2 , and entropy, where smaller AIC and BIC values and larger adjusted ρ^2 and entropy indicate better fit and class separation (Nylund et al., 2007). We selected a six-class solution on the basis of fit statistics and interpretability. In the seven- and eight-class solutions, one class consistently accounted for fewer than 6% of the sample, reducing robustness for segment-level analysis.

Table 3

Latent Class Model Fit Index Comparison.

No. of Classes	Npar	LL	L^2	AIC	BIC	Adj. ρ^2	Entropy	Class proportions
1	11	-20,474	0	40,970	41,056	0.148	—	100.0%
2	33	-17,917	5,114	35,901	36,157	0.254	0.911	78.8% / 21.2%
3	55	-17,411	6,126	34,931	35,358	0.274	0.759	56.0% / 23.4% / 20.6%
4	77	-17,071	6,806	34,296	34,895	0.287	0.758	47.7% / 21.9% / 21.6% / 8.8%
5	99	-16,771	7,406	33,740	34,510	0.299	0.723	25.8% / 22.9% / 21.5% / 21.1% / 8.7%
6	121	-16,538	7,872	33,318	34,258	0.308	0.730	22.7% / 20.8% / 18.5% / 16.6% / 12.8% / 8.7%
7	143	-16,413	8,122	33,113	34,224	0.312	0.736	22.2% / 18.3% / 16.5% / 15.9% / 12.6% / 8.8% / 5.6%
8	165	-16,308	8,332	32,947	34,229	0.316	0.728	20.4% / 16.1% / 16.0% / 15.1% / 12.8% / 7.9% / 5.8% / 5.8%

¹ $L^2 = -2(LL_{lc} - LL_{model})$.² Adj. ρ^2 = adjusted rho-squared vs. observed shares (constants model).³ Entropy = $1 - H/H_{max}$, where $H = -\sum \pi_{ik} \log(\pi_{ik})$. Values closer to 1 indicate cleaner class separation.

5.2.2. WTP and Class Profiles

Estimated model coefficients are presented in Table 4, with **Class 1** serving as the reference class. WTP estimates and class profile characteristics are reported in Table 5, with additional metadata on opt-out behavior and survey engagement in Table 6. Figure 4 provides a visual summary of WTP patterns and segment profiles across classes. Class membership ranges from 8.7% to 22.7% of the sample. We describe each class below, organized to highlight contrasts in battery quality sensitivity, price responsiveness, market readiness, and segment-specific market strategies.

Class 1: Opt-Out Dominant, BEV-Skeptical Consumers ($n=252$, 8.7%).

Class 1 is defined primarily by systematic opt-out behavior: 75.4% of members selected the no-choice option in all six tasks, and an additional 19.4% did so in five of six. As a result, most WTP estimates for vehicle attributes are statistically indistinguishable from zero, with the exception of a moderate disutility for range loss in the 5–12% segment (\$1.6k* per percentage point) and significant penalties for both pack replacement (\$6.4k*) and cell replacement (\$7.3k*). These patterns suggest that respondents who systematically avoid the BEV market retain particular concerns about battery quality.

Class 1 represents consumers who are not yet viable participants in the used BEV market. The majority (70.4%) strongly disagree that they are likely to purchase a used BEV. They are the oldest class on average (47.1 years old), the least risk-tolerant (24.0%), and the least environmentally aware. Their household fleets are almost exclusively internal combustion engine vehicles (ICEVs), with 89.3% of households reporting no EVs and 93.3% driving an ICEV as their primary vehicle. A majority (68.2%) intend their next vehicle to be an SUV. EV-related knowledge is low (68.5% correctly answer the EV knowledge question; only 10.6% are aware of federal EV subsidies), while range anxiety is the highest of any class (87.2%). Attitudes toward refurbished batteries are notably negative: only 30.5% agree that they are environmentally beneficial, and just 9.1% disagree that they are functionally inferior. Infrastructure readiness is also limited: only 32.8% have home electrical outlet access, and only 24.7% report a neighbor who owns or leases a BEV or PHEV. Overall, this class is largely outside the addressable market for used BEVs in the near term.

Class 3: Range-Focused, EV-Engaged Consumers ($n=485$, 16.6%).

Class 3 stands in sharp contrast to Class 1 and is described next to highlight the differences. It is distinguished by an exceptionally high WTP for range: \$38.7k*** per 100 miles for vehicles with 40–130 miles of range, declining to \$20.8k*** (130–200 miles) and \$25.5k*** (200+ miles). This convex pattern suggests that consumers in this class place a higher premium either on lower-range vehicles achieving a minimum viable range threshold (which may reflect their interest in a cost-effective secondary vehicle intended primarily for local trips), or on vehicles with very high ranges (which offer flexibility and reliability for long-distance travel). Consistent with this interpretation, members of this class report the highest average range for their current primary vehicle (321.3 miles), suggesting relatively demanding travel needs and greater sensitivity to vehicle range limitations. Their range loss aversion is moderate and uniformly significant across all degradation segments (\$0.9k***, \$0.7k***, \$0.5k***). Notably, WTP for refurbishment is negligible and statistically insignificant, suggesting that this class focuses more on current and future performance than on battery refurbishment history.

Class 3 represents the most EV-engaged segment in the sample. They hold the most favorable attitudes toward refurbished batteries (65.5% agree they are environmentally beneficial; 35.5% disagree that they are functionally inferior). Their households show the lowest rate of ICEV-only composition (74.1%), and they are the most likely class to prefer a car over an SUV (57.2%). They have the highest electrical outlet access (57.5%), along with the strongest EV knowledge (81.1%) and subsidy awareness (31.9%). They have the highest household income (\$97.1k) and the highest employment rate (71.5%), and are the most risk-tolerant of all classes (39.0% agree with a risk-taking characterization).

For this segment, auto dealers can filter low-range inventory aggressively or provide clear range upgrade pathways. Their minimal disutility for refurbishment history, combined with high income, high EV knowledge, and positive attitudes toward refurbished batteries, suggests they are open to technically sound refurbishments provided the resulting vehicle meets their range threshold. Performance-oriented marketing that leads with current range and degradation trajectory, rather than battery service history, is most likely to resonate.

Class 2: Battery Health Attentives ($n=661$, 22.7%).

Class 2 is the largest class, which is uniquely defined by their battery health sensitivity. Range loss aversion is the highest across classes and uniform at approximately \$2.3k*** per percentage point across all three degradation segments. Both refurbishment types also generate meaningful disutility (\$2.7k* for pack replacement; \$3.8k** for cell replacement). Their range WTP is consistent across the three piecewise segments (\$4.6k, \$4.6k, and \$5.3k**), indicating a steady

marginal valuation of additional range without a strong threshold effect, though only the highest tier reaches statistical significance.

Members of Class 2 have an average household income of \$91.3k and are relatively young (36.5 years). They demonstrate above-average EV knowledge (76.2%), moderate electrical outlet access (46.0%), and a high rate of EV-owning neighbors (43.5%). This class represents well-informed consumers who are not necessarily purchasing range; rather, they are purchasing battery reliability at the point of sale and protection against future performance loss.

Their valuation structure is well suited to transparent, attribute-based pricing. Financing products that tie interest rates or warranty terms to battery SOH readings, or that offer extended coverage contingent on passing a minimum degradation threshold, could be particularly attractive to this segment. Their strong aversion to range loss (\$2.3k per percentage point across all degradation levels) also suggests that short-term ownership scenarios with rapid degradation are significantly discounted, pointing to the value of long-term degradation forecasts in the listing description.

Class 5: Multi-Attribute Maximizers ($n=606$, 20.8%).

Class 5 exhibits statistically significant evaluations across all vehicle attributes. Like Class 3, this class places a high premium on crossing the minimum viable range threshold, with still-substantial but diminishing WTP at higher range levels (\$11.8k***, \$9.5k***, \$9.5k***). Range loss aversion is uniformly significant at \$0.8k*** per percentage point across the lower two degradation segments, tapering at the highest degradation level (\$0.4k*). Refurbishment disutility is among the highest of the attribute-sensitive classes (\$5.8k*** for pack replacement; \$6.1k*** for cell replacement).

What further distinguishes Class 5 is its opt-out pattern: only 5.2% of members never opt out, while the plurality opt out two (29.0%) or three (27.2%) times out of six tasks. Class 5 members exhibit high range anxiety (83.4%) and are particularly averse to battery refurbishment (only 18.8% disagree that refurbished batteries are functionally inferior). They also report average household incomes (\$86.9k) and lower risk tolerance. These characteristics lead them to make cautious, holistic evaluations across all vehicle attributes simultaneously. The uniform magnitude and significance of their attribute benefits and penalties mean that no single attribute can easily offset large deficits elsewhere. Combined with their high opt-out frequency, this suggests that when a choice set presents no option that reaches a minimal quality threshold across all dimensions at once, respondents in this class choose to opt out rather than settle for the least-bad alternative.

This segment is not amenable to price discounting as a close tactic: if a vehicle does not clear the quality bar on all relevant dimensions, a lower price is unlikely to convert. Product curation and clear multi-dimensional disclosure presenting range, degradation rate, and refurbishment status together are more effective approaches for this group.

Class 6: Budget-constrained, Low-WTP Consumers ($n=373$, 12.8%).

Class 6 has the largest price coefficient in absolute value (-4.017^{***}), more than twice the magnitude of the next-most price-sensitive class, suggesting they are budget-constrained buyers. Range WTPs are modest but statistically significant at lower range segments (\$2.8k** for 40–130 miles; \$1.8k* for 130–200 miles), and range loss aversion is marginal (\$0.2k** and \$0.2k*** in the upper degradation segments). WTP for refurbishment is negligible and statistically insignificant, suggesting that battery service history is largely immaterial in this group's purchasing calculus.

Class 6 members have the lowest average household income (\$70.0k) and the lowest next-vehicle budget (\$19.7k). They have the lowest homeownership rate (43.0%) and the highest share of renters (48.8%), the fewest household vehicles (1.9 on average), and the highest share of ICEV-only households (87.1%). Despite their strong price sensitivity, 91.2% of Class 6 members never select the no-choice option, indicating their interest in used vehicles, but their WTP is dominated by price rather than battery quality.

For this class, affordable pricing strategies and accessible financing, including buy-here-pay-here programs, certified pre-owned tiers with minimum quality floors, or OEM-backed affordability programs, are most relevant for serving this segment. Expanding used BEV access to lower-income, renter households hinges largely on whether vehicles in the \$15k–\$20k range can be made commercially viable.

Class 4: Price-Insensitive, Mixed-Motivation Evaluators ($n=539$, 18.5%).

The most prominent feature of this class is its exceptionally low price coefficient in absolute value (-0.144^{**}), indicating substantially lower price sensitivity than all other classes; as a result, all calculated WTP values are ratio-inflated. We suspect this class encompasses several subsets of respondents with different motivations who nonetheless produce similar WTP patterns.

Class 4 exhibits the largest disutility for battery refurbishment of any class: \$26.6k** for pack replacement and \$32.6k** for cell replacement. One contributing factor may be that Class 4 has the highest share of respondents who

1 received the extended information treatment (52.7%), which included more details about battery maintenance costs
2 and warranty coverage. This information may have heightened respondents' attention to the refurbishment attribute
3 and amplified their concern about non-original battery conditions.

4 At the same time, Class 4 exhibits characteristics commonly associated with experienced EV consumers. Class 4
5 has the highest rates of current BEV ownership in their household (13.5%), PHEV or HEV ownership (25.3%), and
6 EV-owning neighbors (50.9%). They report the lowest range anxiety (67.0%) and the highest risk-taking propensity
7 (51.9%), and they have set the highest budget for their next vehicle (\$29.3k). These patterns suggest that at least some
8 respondents in this class are highly committed to EV technology and may view many of the battery-health attributes
9 included in the experiment as less relevant to their purchase decisions.

10 A third explanation relates to respondent engagement. Despite detailed attribute descriptions and explanatory
11 materials, some respondents may not have fully understood the battery-health information presented in the choice
12 tasks. Mileage appears to be the only attribute that generated a large and statistically significant valuation (\$10.0k**
13 per 10,000 miles), whereas most other attributes produced imprecise estimates. This pattern is consistent with the
14 possibility that some respondents focused on the most familiar vehicle characteristic while devoting limited attention
15 to the remaining attributes. Such behavior can generate unstable parameter estimates and exaggerated WTP values that
16 do not necessarily reflect underlying preferences. Supporting this interpretation, this class has a short completion time
17 (on average 81 seconds across six tasks). This raises the possibility that some members of this class did not carefully
18 evaluate all choice attributes, potentially contributing to the imprecise estimates for the majority of the attributes.

19 Taken together, Class 4 may represent a mixture of refurbishment-averse consumers, experienced EV buyers, and
20 attribute-disengaged respondents. Although these groups likely differ in their underlying motivations, they generate a
21 common empirical pattern characterized by very large negative valuations for battery refurbishment alongside limited
22 sensitivity to most other battery-health attributes. As a result, the class is best interpreted cautiously, with consideration
23 given to both information-treatment effects and potential heterogeneity in respondent engagement.



Figure 4: WTP and class profile summary.

Table 4
LCCM model results

	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Vehicle Attributes						
No-Choice Option (opt-out)	-2.577 (1.344).	-9.819 (0.813)***	-1.968 (0.75)**	-3.491 (0.457)***	-2.513 (0.387)***	-13.847 (1.661)***
Mileage (10,000 miles)	-0.472 (0.248).	-0.3 (0.041)***	-0.31 (0.058)***	-0.143 (0.03)***	-0.22 (0.031)***	-0.229 (0.055)***
BEV Electric Range (Year 3): <130 miles	0.114 (1.326)	0.468 (0.281).	5.742 (0.417)***	0.177 (0.225)	1.426 (0.365)***	1.128 (0.399)**
BEV Electric Range (Year 3): 130-200 miles	1.884 (1.383)	0.472 (0.245).	3.089 (0.471)***	-0.145 (0.176)	1.14 (0.22)***	0.707 (0.357)*
BEV Electric Range (Year 3): 200+ miles	1.347 (0.61)*	0.545 (0.189)**	3.78 (0.733)***	0.198 (0.116).	1.145 (0.166)***	0.794 (0.507)
Range Loss Rate: <12%	-0.237 (0.072)***	-0.236 (0.029)***	-0.129 (0.024)***	0.004 (0.015)	-0.093 (0.017)***	-0.084 (0.03)**
Range Loss Rate: 12%-24%	-0.044 (0.063)	-0.239 (0.019)***	-0.102 (0.016)***	-0.018 (0.01).	-0.091 (0.013)***	-0.069 (0.02)***
Range Loss Rate: 24%+	0.035 (0.087)	-0.22 (0.03)***	-0.069 (0.019)***	-0.02 (0.01)*	-0.047 (0.019)*	-0.067 (0.02)***
Battery Refurbishment: Pack Replace	-0.932 (0.431)*	-0.281 (0.126)*	-0.08 (0.159)	-0.384 (0.105)***	-0.695 (0.106)***	0.041 (0.166)
Battery Refurbishment: Cell Replace	-1.074 (0.417)*	-0.385 (0.127)**	-0.099 (0.145)	-0.47 (0.102)***	-0.732 (0.11)***	0.109 (0.165)
Purchase Price (10,000 USD)	-1.467 (0.652)*	-1.025 (0.118)***	-1.484 (0.287)***	-0.144 (0.054)**	-1.206 (0.1)***	-4.017 (0.363)***
Active Indicators						
ASC	0	0.867 (0.5).	-0.272 (0.552)	2.641 (0.493)***	1.379 (0.539)*	1.28 (0.577)*
Perceived EV Range Anxiety: Agree	0	-0.021 (0.257)	-0.131 (0.272)	-0.734 (0.251)**	-0.118 (0.269)	-0.513 (0.283).
Risk Taking Propensity: Agree	0	0.692 (0.2)***	0.387 (0.214).	0.893 (0.204)***	0.042 (0.207)	0.001 (0.242)
Household Income (10,000 USD)	0	0.011 (0.016)	0.021 (0.018)	0.003 (0.017)	0 (0.018)	-0.048 (0.022)*
Electrical Outlet Access	0	0.391 (0.191)*	0.812 (0.205)***	0.525 (0.197)**	0.305 (0.195)	0.423 (0.223).
EV Battery Knowledge: Yes	0	0.146 (0.194)	0.307 (0.225)	-0.183 (0.202)	0.284 (0.197)	0.009 (0.223)
EV Battery Environmentally Positive: Agree	0	1.209 (0.189)***	1.295 (0.211)***	0.808 (0.196)***	0.829 (0.191)***	1.186 (0.217)***
EV Battery Functionally Negative: Disagree	0	0.987 (0.278)***	1.547 (0.284)***	0.822 (0.288)**	0.745 (0.29)*	1.169 (0.303)***
Current Primary Household Vehicle Fuel Type: ICEV	0	-0.629 (0.268)*	-0.592 (0.279)*	-1.227 (0.261)***	-0.596 (0.268)*	-0.015 (0.332)
Current Primary Vehicle Typical Range (miles)	0	-0.173 (0.091).	0.076 (0.098)	-0.325 (0.094)***	-0.113 (0.096)	-0.186 (0.101).
SUV/Crossover as the Next Vehicle	0	-0.203 (0.191)	-0.984 (0.273)***	-0.52 (0.198)**	-0.531 (0.191)**	-0.736 (0.222)***

Est. (SE)[Sig.]

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1.

Table 5
WTP and Characteristics by Class

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
WTP (*1000 USD) for Vehicle Attributes						
No-Choice Option (opt-out)	-\$17.6.	-\$95.8***	-\$13.3**	-\$242.4**	-\$20.8***	-\$34.5***
Mileage (10,000 miles)	-\$3.2.	-\$2.9***	-\$2.1***	-\$10.0**	-\$1.8***	-\$0.6***
BEV Range: 40-130 mi segment (per 100 miles)	\$0.8	\$4.6.	\$38.7***	\$12.3	\$11.8***	\$2.8**
BEV Range: 130-200 mi segment (per 100 miles)	\$12.8	\$4.6.	\$20.8***	-\$10.1	\$9.5***	\$1.8*
BEV Range: 200+ mi segment (per 100 miles)	\$9.2*	\$5.3**	\$25.5***	\$13.8.	\$9.5***	\$2.0
Range Loss Rate: 5-12% (per %)	-\$1.6*	-\$2.3***	-\$0.9***	\$0.3	-\$0.8***	-\$0.2**
Range Loss Rate: 12-24% (per %)	-\$0.3	-\$2.3***	-\$0.7***	-\$1.3.	-\$0.8***	-\$0.2***

Table 5
WTP and Characteristics by Class (*continued*)

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
Range Loss Rate: 24%+ (per %)	\$0.2	-\$2.1***	-\$0.5***	-\$1.4*	-\$0.4*	-\$0.2***
Battery Refurbishment: Pack Replace	-\$6.4*	-\$2.7*	-\$0.5	-\$26.6**	-\$5.8***	\$0.1
Battery Refurbishment: Cell Replace	-\$7.3*	-\$3.8**	-\$0.7	-\$32.6**	-\$6.1***	\$0.3
Active Indicators						
Risk-taking Propensity: Agree: no	76.0%	56.0%	61.0%	48.1%	71.3%	72.4%
Risk-taking Propensity: Agree: yes	24.0%	44.0%	39.0%	51.9%	28.7%	27.6%
Perceived EV Range Anxiety: Agree: no	12.8%	17.0%	19.4%	33.0%	16.6%	22.0%
Perceived EV Range Anxiety: Agree: yes	87.2%	83.0%	80.6%	67.0%	83.4%	78.0%
EV Battery Environmentally Positive: Agree: no	69.5%	37.4%	34.5%	46.4%	47.7%	39.6%
EV Battery Environmentally Positive: Agree: yes	30.5%	62.6%	65.5%	53.6%	52.3%	60.4%
EV Battery Functionally Negative: Disagree: no	90.9%	76.9%	64.5%	78.5%	81.2%	72.1%
EV Battery Functionally Negative: Disagree: yes	9.1%	23.1%	35.5%	21.5%	18.8%	27.9%
Household Income (1000 USD)	85.5	91.3	97.1	89.3	86.9	70.0
EV Knowledge: no	31.5%	23.8%	18.9%	28.4%	23.5%	30.7%
EV Knowledge: yes	68.5%	76.2%	81.1%	71.6%	76.5%	69.3%
Electrical Outlet Access: no	63.0%	47.8%	37.2%	42.3%	51.8%	52.3%
Electrical Outlet Access: not_sure	4.1%	6.3%	5.3%	5.8%	6.4%	7.8%
Electrical Outlet Access: yes	32.8%	46.0%	57.5%	51.9%	41.8%	39.8%
Primary Vehicle Typical Range (miles)	306.8	292.2	321.3	267.6	297.4	282.2
Household Vehicle Fuel Composition: has_bev	1.7%	7.5%	9.2%	13.5%	7.5%	3.4%
Household Vehicle Fuel Composition: has_phev_hev	7.6%	16.1%	16.6%	25.3%	13.3%	9.5%
Household Vehicle Fuel Composition: icev_only	89.3%	76.2%	74.1%	61.2%	79.0%	87.1%
Household Vehicle Fuel Composition: other	1.4%	0.1%	0.0%	0.0%	0.2%	0.0%
Next Vehicle Type: SUV: no	31.8%	38.9%	57.2%	48.3%	46.3%	53.3%
Next Vehicle Type: SUV: yes	68.2%	61.1%	42.8%	51.7%	53.7%	46.7%
Inactive Indicators						
EV Subsidy Knowledge: no	89.4%	73.6%	68.1%	70.9%	78.0%	79.1%
EV Subsidy Knowledge: yes	10.6%	26.4%	31.9%	29.1%	22.0%	20.9%
Neighbor Owns/Leases a BEV/PHEV: no	46.9%	29.6%	26.0%	28.5%	31.8%	33.9%
Neighbor Owns/Leases a BEV/PHEV: not_sure	28.3%	26.9%	26.8%	20.6%	30.3%	31.2%
Neighbor Owns/Leases a BEV/PHEV: yes	24.7%	43.5%	47.2%	50.9%	37.9%	34.9%
Age	47.1	36.5	37.5	37.4	40.9	37.7
Gender: female	67.3%	65.1%	52.6%	57.9%	63.1%	60.1%
Gender: male	31.3%	32.7%	44.9%	41.3%	34.4%	36.0%
Gender: other	1.4%	1.9%	2.5%	0.8%	2.5%	3.2%
Gender: prefer_not_answer	0.0%	0.3%	0.0%	0.1%	0.0%	0.7%
Ethnicity: hispanic	5.3%	11.6%	9.8%	15.5%	11.5%	11.6%
Ethnicity: non-hispanic	94.7%	88.4%	90.2%	84.5%	88.5%	88.4%
Race: african_american_only	7.1%	17.9%	14.4%	22.3%	11.4%	11.7%
Race: other	10.4%	15.6%	14.5%	18.5%	18.1%	15.9%
Race: white_only	82.5%	66.5%	71.2%	59.2%	70.5%	72.4%
Education Level: bachelor	31.9%	36.8%	38.4%	32.9%	36.1%	31.7%
Education Level: graduate	17.0%	20.2%	20.4%	20.4%	19.6%	17.9%
Education Level: high_school	18.6%	12.6%	11.9%	16.8%	11.4%	16.0%
Education Level: prefer_not_answer	0.0%	0.2%	0.0%	0.1%	0.5%	0.5%
Education Level: some_college	32.6%	30.2%	29.3%	29.8%	32.4%	33.9%
Student Status: non-student	92.4%	84.1%	84.6%	84.0%	86.0%	80.6%
Student Status: prefer_not_answer	0.7%	1.1%	0.8%	0.9%	1.8%	1.9%
Student Status: student	6.9%	14.8%	14.6%	15.1%	12.1%	17.5%
Employment Status: full_time	38.2%	47.2%	48.4%	46.9%	43.1%	38.6%
Employment Status: not_employed	36.5%	26.0%	27.7%	29.4%	33.1%	33.5%
Employment Status: part_time	24.6%	25.7%	23.1%	22.8%	21.9%	26.1%
Employment Status: prefer_not_answer	0.7%	1.1%	0.8%	0.9%	1.8%	1.9%
Household Size	2.8	3.1	2.9	3.3	2.9	2.8
Household Tenure: other	6.1%	6.1%	5.8%	4.7%	4.9%	7.2%
Household Tenure: own	58.3%	49.6%	54.4%	49.5%	52.5%	43.0%

Table 5
WTP and Characteristics by Class (*continued*)

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
Household Tenure: prefer_not_answer	1.6%	1.2%	1.8%	1.8%	1.6%	1.0%
Household Tenure: rent	34.1%	43.0%	38.1%	43.9%	41.0%	48.8%
Household Type: apart	17.2%	24.1%	21.6%	26.8%	21.9%	27.6%
Household Type: other	4.7%	3.2%	3.1%	3.2%	4.1%	3.9%
Household Type: sf_attached	9.2%	11.9%	9.2%	12.4%	10.6%	10.3%
Household Type: sf_detached	68.9%	60.9%	66.0%	57.6%	63.5%	58.2%
Household Vehicle Count	2.0	2.1	2.1	2.1	2.0	1.9
Primary Vehicle Fuel Type: bev	0.9%	3.1%	4.7%	6.1%	4.3%	1.4%
Primary Vehicle Fuel Type: icev	93.3%	85.3%	84.9%	72.0%	87.4%	91.6%
Primary Vehicle Fuel Type: phev_hev	5.8%	11.6%	10.4%	21.9%	8.3%	6.9%
Next Vehicle Budget (1000 USD)	24.3	27.8	26.9	29.3	25.2	19.7
Likelihood of buying new BEV: neutral	5.2%	10.7%	11.3%	14.8%	10.0%	10.0%
Likelihood of buying new BEV: somewhat_agree	5.3%	16.8%	16.7%	22.3%	13.1%	12.5%
Likelihood of buying new BEV: somewhat_disagree	10.6%	20.4%	22.2%	16.9%	16.4%	19.1%
Likelihood of buying new BEV: strongly_agree	1.7%	7.4%	5.8%	11.1%	4.6%	3.3%
Likelihood of buying new BEV: strongly_disagree	77.2%	44.6%	44.0%	34.9%	55.9%	55.2%
Likelihood of buying used BEV: neutral	5.5%	12.6%	13.2%	16.7%	12.3%	13.6%
Likelihood of buying used BEV: somewhat_agree	6.3%	26.2%	26.2%	26.7%	18.9%	23.4%
Likelihood of buying used BEV: somewhat_disagree	14.7%	20.5%	24.1%	18.0%	19.7%	22.0%
Likelihood of buying used BEV: strongly_agree	3.2%	10.5%	10.1%	11.1%	8.0%	7.5%
Likelihood of buying used BEV: strongly_disagree	70.4%	30.2%	26.4%	27.6%	41.2%	33.5%
Climate Concern: no	78.3%	56.5%	57.6%	62.4%	59.8%	58.4%
Climate Concern: yes	21.7%	43.5%	42.4%	37.6%	40.2%	41.6%

Table 6
Metadata Summary by Class

	Class 1 (n=252, 8.7%)	Class 2 (n=661, 22.7%)	Class 3 (n=485, 16.6%)	Class 4 (n=539, 18.5%)	Class 5 (n=606, 20.8%)	Class 6 (n=373, 12.8%)
Metadata: Information Treatment						
Battery Information Treatment: no	51.6%	49.3%	50.7%	47.3%	48.3%	50.3%
Battery Information Treatment: yes	48.4%	50.7%	49.3%	52.7%	51.7%	49.7%
Metadata: Opt-out Count						
Opt-out: 0 times	0.0%	86.4%	85.7%	85.0%	5.2%	91.2%
Opt-out: 1 time	0.0%	12.0%	10.7%	12.2%	18.6%	7.4%
Opt-out: 2 times	0.0%	1.6%	2.9%	2.7%	29.0%	1.1%
Opt-out: 3 times	0.4%	0.1%	0.6%	0.1%	27.2%	0.3%
Opt-out: 4 times	4.8%	0.0%	0.0%	0.0%	15.3%	0.0%
Opt-out: 5 times	19.4%	0.0%	0.0%	0.0%	4.1%	0.0%
Opt-out: 6 times	75.4%	0.0%	0.0%	0.0%	0.6%	0.0%
Metadata: Survey Duration						
Battery DCE Section: Q1	22.0	25.0	27.0	22.0	27.0	25.0
Battery DCE Section: Q2	9.0	15.0	16.0	14.0	16.0	15.0
Battery DCE Section: Q3	7.0	14.0	15.0	12.0	14.0	13.0
Battery DCE Section: Q4	7.0	13.0	15.0	12.0	13.0	13.0
Battery DCE Section: Q5	6.0	12.0	13.0	11.0	12.0	12.0
Battery DCE Section: Q6	6.0	12.0	13.0	11.0	11.0	11.0
Battery DCE Section: Total	63.0	99.0	107.0	93.0	102.0	97.0
Full Survey	896.0	924.0	907.0	952.0	894.0	842.0

5.3. Results of Opt-Out Analysis

The thematic network for the reasons for opt-out ($n = 209$) is plotted in Figure 5. The network visualization reveals 11 clusters of opt-out reasons (ranked by frequency below): general disinterest in BEVs, practical barriers (economic barriers, charging inconvenience, range anxiety, battery concerns), psychological factors (gas vehicle enthusiasm, EV distrust, used vehicle distrust, environmental concerns), and limited knowledge on BEVs.

The most prevalent reason is *general disinterest in BEVs* ($n=54$), expressed without reference to any concrete barrier (e.g., “I just don’t want one”). *Economic Barriers* ($n = 52$) group four sub-themes: the upfront purchase price, ongoing maintenance and repair expenses (in which battery replacement looms large), operation costs (e.g., electricity cost), and a generic “not worth it” value judgment. *Charging Inconvenience* ($n = 51$) captures three sub-themes: insufficient access to public chargers (especially in rural or remote areas), lack of home charging access (often tied to apartment or rental living situations without a private outlet), and the time required to recharge relative to refueling a gasoline vehicle. *Range Anxiety* ($n = 42$) encompasses two sub-themes: insufficient range for daily driving needs, and the fear of being stranded on long trips beyond a single charge. *Battery Concerns* ($n = 40$) are anchored in three sub-themes: capacity degradation and lifespan loss, safety risks such as fire or crash damage to lithium-ion cells, and cold-weather performance loss.

Gas Vehicle Enthusiasm ($n = 43$) reflects a positive preference for internal combustion engine vehicles, split between an explicit love of gas engines (sound, range, refueling speed) and the absence of certain features in BEVs such as manual transmissions. *EV Distrust* ($n = 33$) captures general skepticism of the technology, comprising a sub-theme of generic “no faith” statements and a related concern that the technology is still immature or not ready for mainstream adoption. *Used Vehicle Distrust* ($n = 15$) refers specifically to skepticism toward refurbished or used BEV batteries; respondents in this group often reported they would only consider a new BEV, if any. *Environmental Concerns* ($n = 9$) comprise a broader “green claim” skepticism (e.g., that BEVs are not actually better for the environment than gasoline vehicles) and concerns about end-of-life battery disposal and recycling. Finally, *Limited Knowledge on BEVs* ($n = 4$) captures the smallest share who explicitly admitted insufficient familiarity to make an informed choice.

Following Krishna (2021), we further examine the interdependencies among barriers, that is, the extent to which the same respondent simultaneously cites concerns from different parent themes. These interdependencies are shown as dashed lines in Figure 5, with the count labeled near each line. The strongest interdependency forms a practical-infeasibility cluster linking charging and range: respondents who mention insufficient public chargers also mention long-trip range insufficiency ($n = 13$), and charging time co-occurs with insufficient daily range ($n = 10$). Battery-related concerns bridge into adjacent clusters: battery degradation co-occurs with used-vehicle distrust ($n = 6$), reflecting that the used BEVs distrust could be associated with battery replacement. Gas engine preference bridges to both psychological and practical barriers, including insufficient public chargers ($n = 10$), general EV distrust ($n = 27$), and insufficient daily range ($n = 7$). These overlaps indicate that addressing any single barrier (e.g., expanding public charging infrastructure) could plausibly alleviate several adjacent concerns simultaneously, consistent with the cascading-effect pattern reported by Krishna (2021).

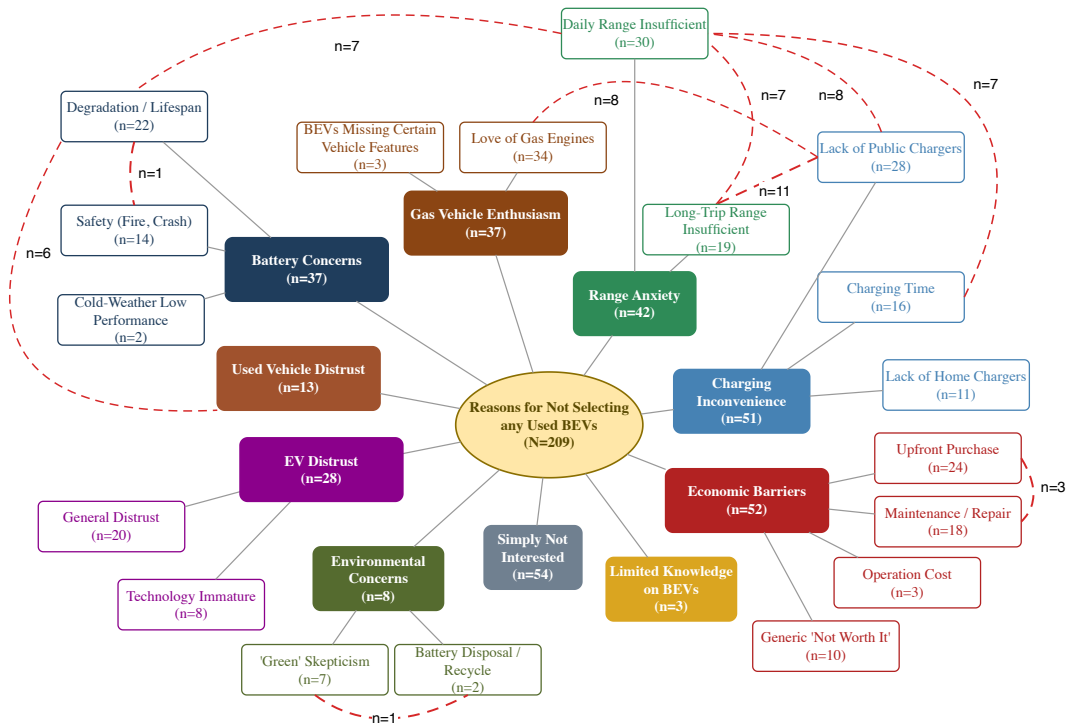


Figure 5: Thematic network and interdependencies of opt-out reasons.

Regarding the impact of the information treatment on opt-out, the opt-out rate in the total sample is 6.81% (=103/1513) for the basic-information group and 6.80% (=106/1559) for the extended-information group. The statistical test confirms that the information treatment does not affect the overall opt-out rate ($\chi^2 \approx 0$, $p = 1.00$).

However, the following two bar plots (Figure 6 and Figure 7) indicate that while some themes are common across both treatment groups, there are differences in the prevalence of specific concerns. Note that the sub-theme-level comparisons reveal more nuanced patterns but should be interpreted cautiously due to the small sample sizes of several sub-themes. Because the chi-square approximation is unreliable in this setting, we report Fisher's exact p -values; even so, statistical power to detect realistic effects is low. The findings below should therefore be read as descriptive and hypothesis-generating rather than confirmatory, and larger- N replications will be needed to verify the treatment-channel mechanism we propose. Nevertheless, we believe that some of the observations are economically meaningful in direction.

Only one treatment difference reaches statistical significance: *general disinterest in BEVs* is markedly more common among respondents receiving extended information (32.1%) than among basic-information respondents (19.4%, $p = 0.037$). Several other themes and sub-themes show economically meaningful directional differences that do not reach statistical significance. For instance, extended-information respondents are more likely to cite the *general EV distrust* sub-theme (11.3% vs. 7.8%), the *generic "not worth it"* sub-theme (5.7% vs. 3.9%), and the *battery replacement cost* sub-theme (5.7% vs. 3.9%), the last of which is particularly highlighted in the information treatment. In contrast, basic-information respondents are more likely to cite specific barriers across most of the sub-themes listed in Figure 7, especially *economic barriers* (29.1% vs. 20.8%), *charging inconvenience* (28.2% vs. 20.8%), and *range anxiety* (23.3% vs. 17.0%).

In sum, these differences are consistent with a channel-substitution interpretation in which extended battery information does not change the overall opt-out rate but redirects how the reasons are articulated. Extended-information respondents are significantly more likely to consolidate reluctance into generic "not interested" or "distrust" responses. Basic-information respondents, lacking that specific information, default to broader practical concerns such as upfront cost, lack of public charging access, and range insufficiency.

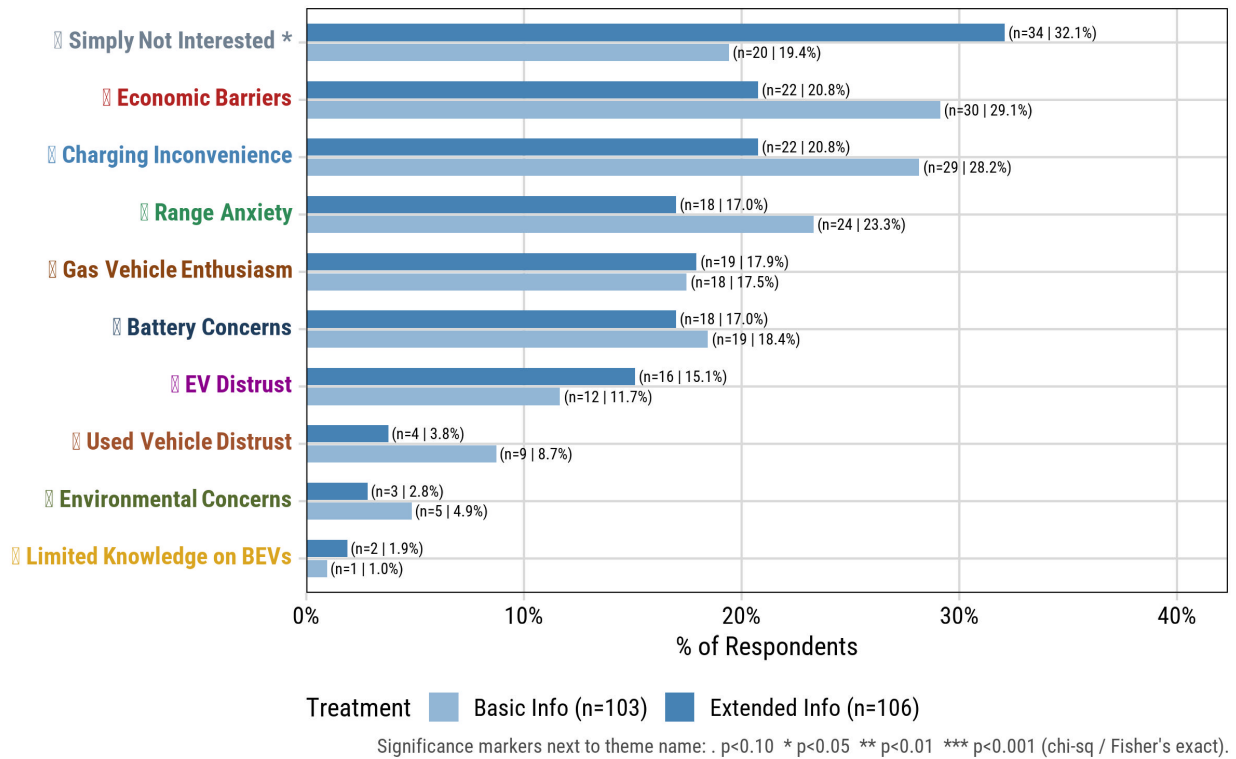


Figure 6: Self-reported opt-out reasons by information treatment group (by themes).

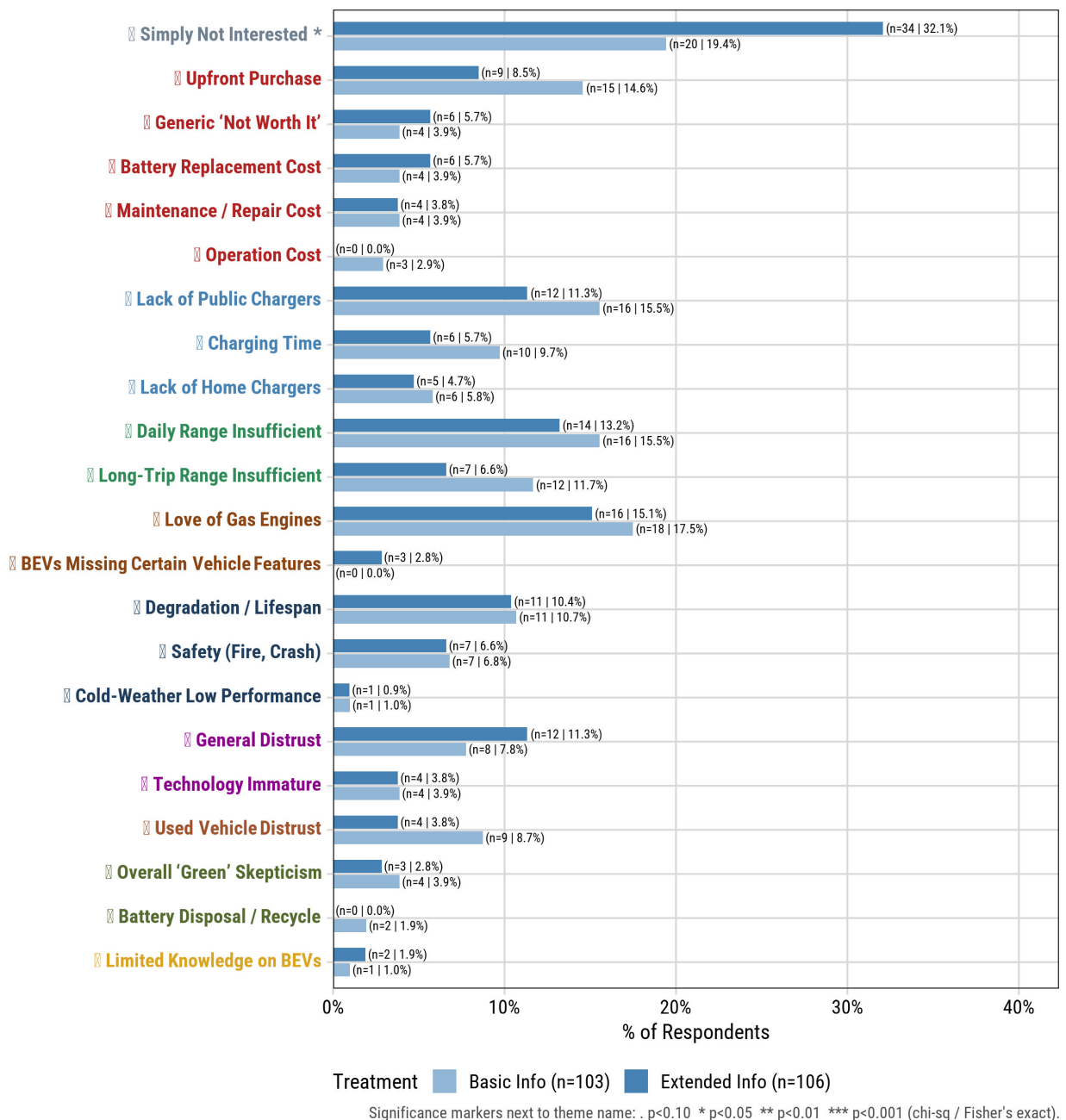


Figure 7: Self-reported opt-out reasons by information treatment group (by sub-themes).

6. Conclusion

The used BEV market is expanding rapidly, yet persistent information asymmetry around battery health continues to suppress consumer confidence and limit market efficiency. This paper provides the first WTP estimates for battery health information in the U.S. used BEV market, based on a national DCE with 3,072 respondents and a jointly estimated MXL and six-class LCCM.

At the aggregate level, consumers value electric range, penalize mileage and battery degradation, and discount refurbishment history. The MXL estimates confirm that preference heterogeneity is substantial. The LCCM reveals that consumers can be organized into six segments that differ markedly in their sensitivity to battery quality signals,

their price responsiveness, and their market engagement. **Class 1** (8.7%) is defined by systematic opt-out behavior and represents consumers who are largely outside the addressable used BEV market, with low EV knowledge, high range anxiety, and predominantly ICEV households. **Class 2** (22.7%), the largest class, is characterized by the strongest and most uniform range loss aversion and meaningful disutility for both refurbishment types; these well-informed consumers are effectively purchasing battery reliability rather than range per se. **Class 3** (16.6%) is the most EV-engaged segment and places an exceptionally high premium on crossing a minimum viable range threshold. **Class 5** (20.8%) attends simultaneously to range adequacy, degradation trajectory, and refurbishment condition, and opts out selectively when no presented vehicle meets its standards across all dimensions at once. **Class 6** (12.8%) is the most price-sensitive class and is composed of budget-constrained buyers who are strongly committed to making a vehicle choice but largely indifferent to battery quality signals. **Class 4** (18.5%) is best interpreted cautiously: its near-zero price sensitivity inflates calculated WTP values, and the class likely encompasses a mix of information-treatment-affected respondents, experienced EV adopters, and attribute-disengaged respondents who share a common pattern of large refurbishment disutility alongside imprecise estimates for most other attributes.

These findings carry a direct message for dealers and OEMs. Vehicles with documented performance indicators (mileage, range) and battery health information (degradation rate and refurbishment history) represent a material pricing-premium opportunity in the majority of the market. It is important to consider the development of standardized battery health certification, analogous in structure and function to vehicle history reports. A certification program that provides independently verified SOH readings, degradation history, and remaining warranty coverage could allow dealers to differentiate inventory and command measurable price premiums from the battery-quality-sensitive majority. Critically, cell and pack replacement are penalized at similar magnitudes despite their technical differences, which indicates that consumers do not currently distinguish between high-quality refurbishment and simple repair. This represents a market opportunity for third-party certification services that validate refurbishment quality and communicate it in terms consumers understand.

The information treatment analysis adds an important qualification: providing extended battery and warranty information does not change overall market (dis)engagement, but it redirects the type of concern driving opt-out behavior. Extended-information respondents are more likely to articulate reluctance as generic disinterest or distrust and as battery-replacement-cost concerns, while basic-information respondents articulate it as more specific practical barriers such as upfront price and infrastructure availability. This finding suggests that disclosure alone is insufficient to build purchase confidence. Credible assurance mechanisms, such as third-party SOH certification, warranty-backed condition guarantees, and transparent degradation forecasts, are needed to give disclosed battery information a positive commercial meaning rather than surfacing a risk without resolution.

Several limitations should be noted. The study relies on stated preferences from a DCE, which may not fully replicate the decision-making dynamics of real vehicle purchases, particularly for consumers with limited prior EV experience. The sample is restricted to individuals who express an intention to purchase a used vehicle within two years, which excludes passive or uninterested populations. The LCCM assumes a finite number of discrete classes; unobserved within-class heterogeneity remains. Future research should examine how these preferences evolve as used BEV supply increases and battery health reporting becomes more standardized, and whether incentive programs aimed at improving information quality in the secondary market translate into measurable shifts in demand.

7. Appendix

CRediT authorship contribution statement

Xiatian Iogansen: Conceptualization, Data curation, Methodology, Software, Formal analysis, Writing - Original Draft, Writing - Review & Editing. **Zain Hoda:** Conceptualization, Data curation, Methodology, Writing - Review & Editing. **Christina Gore:** Conceptualization, Data curation, Methodology, Investigation, Writing - Original draft preparation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Joshua D. Kneifel:** Conceptualization, Data curation, Methodology, Investigation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Sindhu Ranganath:** Writing - Original Draft, Writing - Review & Editing. **John P. Helveston:** Conceptualization, Methodology, Writing - Review & Editing, Supervision.

References

- Attia, P. M., A. Bills, F. Brosa Planella, P. Dechent, G. dos Reis, M. Dubarry, P. Gasper, R. Gilchrist, S. Greenbank, D. Howey, O. Liu, E. Khoo, Y. Preger, A. Soni, S. Sripad, A. G. Stefanopoulou, and V. Sulzer (June 2022). "Review—"Knees" in Lithium-Ion Battery Aging Trajectories". In: *Journal of The Electrochemical Society* 169.6, p. 060517. ISSN: 1945-7111. DOI: 10.1149/1945-7111/ac6d13. URL: <https://dx.doi.org/10.1149/1945-7111/ac6d13> (visited on 07/03/2025).
- Attia, P. M., W. C. Chueh, and S. J. Harris (May 2020). "Revisiting the T0.5 Dependence of SEI Growth". In: *Journal of The Electrochemical Society* 167.9, p. 090535. ISSN: 1945-7111. DOI: 10.1149/1945-7111/ab8ce4. URL: <https://dx.doi.org/10.1149/1945-7111/ab8ce4> (visited on 07/03/2025).
- Aviloo (Sept. 22, 2026). URL: <https://aviloo.com/en-us/> (visited on 06/30/2026).
- Bashash, S., S. J. Moura, J. C. Forman, and H. K. Fathy (Jan. 1, 2011). "Plug-in Hybrid Electric Vehicle Charge Pattern Optimization for Energy Cost and Battery Longevity". In: *Journal of Power Sources* 196.1, pp. 541–549. ISSN: 0378-7753. DOI: 10.1016/j.jpowsour.2010.07.001. URL: <https://www.sciencedirect.com/science/article/pii/S0378775310011390> (visited on 07/03/2025).
- Battery-Tech Network (July 10, 2025). *Battery Testing & Analytics Companies*. URL: <https://battery-tech.net/battery-testing-analytics-companies/> (visited on 06/30/2026).
- Boudway, I. (Dec. 16, 2020). "Batteries For Electric Cars Speed Toward a Tipping Point". In: *Bloomberg.com*. URL: <https://www.bloomberg.com/news/articles/2020-12-16/electric-cars-are-about-to-be-as-cheap-as-gas-powered-models> (visited on 07/03/2025).
- Braun, V. and V. Clarke (Jan. 1, 2006). "Using Thematic Analysis in Psychology". In: *Qualitative Research in Psychology* 3.2, pp. 77–101. ISSN: 1478-0887. DOI: 10.1191/1478088706qp063oa. URL: <https://doi.org/10.1191/1478088706qp063oa> (visited on 06/28/2026).
- Brückmann, G., M. Wicki, and T. Bernauer (Nov. 2021). "Is Resale Anxiety an Obstacle to Electric Vehicle Adoption? Results from a Survey Experiment in Switzerland". In: *Environmental Research Letters* 16.12, p. 124027. ISSN: 1748-9326. DOI: 10.1088/1748-9326/ac3531. URL: <https://dx.doi.org/10.1088/1748-9326/ac3531> (visited on 06/30/2025).
- Canals Casals, L., M. Etxandi-Santolaya, P. A. Bibiloni-Mulet, C. Corchero, and L. Trilla (Oct. 2022). "Electric Vehicle Battery Health Expected at End of Life in the Upcoming Years Based on UK Data". In: *Batteries* 8.10 (10), p. 164. ISSN: 2313-0105. DOI: 10.3390/batteries8100164. URL: <https://www.mdpi.com/2313-0105/8/10/164> (visited on 07/02/2025).
- Canals Casals, L., M. Rodríguez, C. Corchero, and R. E. Carrillo (Dec. 2019). "Evaluation of the End-of-Life of Electric Vehicle Batteries According to the State-of-Health". In: *World Electric Vehicle Journal* 10.4 (4), p. 63. ISSN: 2032-6653. DOI: 10.3390/wevj10040063. URL: <https://www.mdpi.com/2032-6653/10/4/63> (visited on 07/02/2025).
- Canepa, K., S. Hardman, and G. Tal (June 1, 2019). "An Early Look at Plug-in Electric Vehicle Adoption in Disadvantaged Communities in California". In: *Transport Policy* 78, pp. 19–30. ISSN: 0967-070X. DOI: 10.1016/j.tranpol.2019.03.009. URL: <https://www.sciencedirect.com/science/article/pii/S0967070X18303524> (visited on 07/01/2025).
- Chen, C.-F., C. Fu, and Y.-C. Chen (Aug. 2023). "Exploring Tourist Preference for Mobility-as-a-Service (MaaS) – A Latent Class Choice Approach". In: *Transportation Research Part A: Policy and Practice* 174, p. 103750. ISSN: 09658564. DOI: 10.1016/j.tra.2023.103750. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0965856423001702> (visited on 05/18/2026).
- Clarke, W. (2024). *Car Warranty Coverage on an Electric Car Battery*. US News & World Report. URL: <https://cars.usnews.com/cars-trucks/advice/ev-battery-warranty> (visited on 07/02/2025).
- Cox Automotive Inc. (2025). *EV Market Monitor – February 2025*. URL: <https://www.coxautoinc.com/market-insights/ev-market-monitor-february-2025/> (visited on 06/30/2025).
- Dnistran, I. (2024). *EV Batteries Can Outlast A Vehicle's Lifetime With Minimal Degradation, Study Finds*. InsideEVs. URL: <https://insideevs.com/news/733987/ev-batteries-outlast-vehicle-degradation-study/> (visited on 07/02/2025).
- Fernandes, J. (2023). *US: Understanding Consumer Opinion about Used Electric Cars*. URL: <https://business.yougov.com/content/47834-us-understanding-consumer-opinion-about-used-electric-cars> (visited on 06/30/2025).
- Gore, C., X. Iogansen, J. Kneifel, and J. Helveston (2026). "DCI: Consumer Valuation of Battery Health Information in Used Electric Vehicle Markets". In: *NIST* (). URL: <https://www.nist.gov/publications/dci-consumer-valuation-battery-health-information-used-electric-vehicle-markets> (visited on 05/11/2026).
- Greene, W. H. and D. A. Hensher (2003). "A Latent Class Model for Discrete Choice Analysis: Contrasts with Mixed Logit". In: *Transportation Research Part B: Methodological* 37.8, pp. 681–698. URL: <https://ideas.repec.org/a/eee/transb/v37y2003i8p681-698.html> (visited on 05/18/2026).
- Guzek, M., J. Jackowski, R. S. Jurecki, E. M. Szumska, P. Zdanowicz, and M. Żmuda (Jan. 2024). "Electric Vehicles—An Overview of Current Issues—Part 1—Environmental Impact, Source of Energy, Recycling, and Second Life of Battery". In: *Energies* 17.1 (1), p. 249. ISSN: 1996-1073. DOI: 10.3390/en17010249. URL: <https://www.mdpi.com/1996-1073/17/1/249> (visited on 07/01/2025).
- Hagman, J., S. Ritzén, J. J. Stier, and Y. Susilo (Mar. 1, 2016). "Total Cost of Ownership and Its Potential Implications for Battery Electric Vehicle Diffusion". In: *Research in Transportation Business & Management*. Innovations in Technologies for Sustainable Transport 18, pp. 11–17. ISSN: 2210-5395. DOI: 10.1016/j.rtbm.2016.01.003. URL: <https://www.sciencedirect.com/science/article/pii/S2210539516000043> (visited on 07/01/2025).
- Helveston, J. P. (Feb. 7, 2023). "Logitr: Fast Estimation of Multinomial and Mixed Logit Models with Preference Space and Willingness-to-Pay Space Utility Parameterizations". In: *Journal of Statistical Software* 105, pp. 1–37. ISSN: 1548-7660. DOI: 10.18637/jss.v105.i10. URL: <https://doi.org/10.18637/jss.v105.i10> (visited on 06/24/2025).
- Helveston, J. P., E. M. Feit, and J. J. Michalek (2018). "Pooling Stated and Revealed Preference Data in the Presence of RP Endogeneity". In: *Transportation Research Part B: Methodological* 109, pp. 70–89. URL: <https://www.sciencedirect.com/science/article/pii/S0191261516304647> (visited on 06/05/2026).
- Iogansen, X., K. Wang, D. Bunch, G. Matson, and G. Circella (Feb. 1, 2023). "Deciphering the Factors Associated with Adoption of Alternative Fuel Vehicles in California: An Investigation of Latent Attitudes, Socio-Demographics, and Neighborhood Effects". In: *Transportation Research*

Part A: Policy and Practice 168, p. 103535. ISSN: 0965-8564. DOI: 10.1016/j.tra.2022.10.012. URL: <https://www.sciencedirect.com/science/article/pii/S0965856422002701> (visited on 06/10/2026).

Jiao, N. and S. Evans (Jan. 1, 2016). "Business Models for Sustainability: The Case of Second-life Electric Vehicle Batteries". In: *Procedia CIRP*. 13th Global Conference on Sustainable Manufacturing – Decoupling Growth from Resource Use 40, pp. 250–255. ISSN: 2212-8271. DOI: 10.1016/j.procir.2016.01.114. URL: <https://www.sciencedirect.com/science/article/pii/S2212827116001293> (visited on 07/07/2025).

Kamakura, W. A. and G. J. Russell (Nov. 1, 1989). "A Probabilistic Choice Model for Market Segmentation and Elasticity Structure". In: *Journal of Marketing Research* 26.4, pp. 379–390. ISSN: 0022-2437. DOI: 10.1177/002224378902600401. URL: <https://doi.org/10.1177/002224378902600401> (visited on 05/18/2026).

Khaloei, M., T. Zou, and D. MacKenzie (2020). *How Does Charging Network Design Affect Electric Vehicle Adoption?* URL: <https://cpb-us-e1.wpmucdn.com/sites.uw.edu/dist/d/4543/files/2020/07/EV-infrastructure-and-vehicle-choice-final-report.pdf> (visited on 07/01/2025).

Kothari, S. (2024). *Battery Replacement Costs Are Poised To Plunge: 'Cheaper Than Fixing An Engine'*. InsideEVs. URL: <https://insideevs.com/news/742022/battery-replacement-costs-fall-cheaper-than-fixing-engine/> (visited on 03/31/2025).

Krishna, G. (June 1, 2021). "Understanding and Identifying Barriers to Electric Vehicle Adoption through Thematic Analysis". In: *Transportation Research Interdisciplinary Perspectives* 10, p. 100364. ISSN: 2590-1982. DOI: 10.1016/j.trip.2021.100364. URL: <https://www.sciencedirect.com/science/article/pii/S2590198221000713> (visited on 07/01/2025).

Letmathe, P. and M. Suares (Dec. 1, 2017). "A Consumer-Oriented Total Cost of Ownership Model for Different Vehicle Types in Germany". In: *Transportation Research Part D: Transport and Environment* 57, pp. 314–335. ISSN: 1361-9209. DOI: 10.1016/j.trd.2017.09.007. URL: <https://www.sciencedirect.com/science/article/pii/S1361920917304868> (visited on 07/07/2025).

Lim, M. K., H.-Y. Mak, and Y. Rong (Feb. 2015). "Toward Mass Adoption of Electric Vehicles: Impact of the Range and Resale Anxieties". In: *Manufacturing & Service Operations Management* 17.1, pp. 101–119. ISSN: 1523-4614. DOI: 10.1287/msom.2014.0504. URL: <https://pubsonline.informs.org/doi/10.1287/msom.2014.0504> (visited on 06/30/2025).

Market Research Future (2025). *United States Used Car Market Trends & Growth By 2034*. URL: <https://www.marketresearchfuture.com/reports/us-used-car-market-21771> (visited on 07/01/2025).

McFadden, D. and K. Train (Sept. 2000). "Mixed MNL Models for Discrete Response". In: *Journal of Applied Econometrics* 15.5, pp. 447–470. ISSN: 0883-7252, 1099-1255. DOI: 10.1002/1099-1255(200009/10)15:5<447::AID-JAE570>3.0.CO;2-1. URL: [https://onlinelibrary.wiley.com/doi/10.1002/1099-1255\(200009/10\)15:5%3C447::AID-JAE570%3E3.0.CO;2-1](https://onlinelibrary.wiley.com/doi/10.1002/1099-1255(200009/10)15:5%3C447::AID-JAE570%3E3.0.CO;2-1) (visited on 06/05/2026).

Naseri, H., E. O. D. Waygood, Z. Patterson, and B. Wang (Sept. 1, 2024). "Who Is More Likely to Buy Electric Vehicles?" In: *Transport Policy* 155, pp. 15–28. ISSN: 0967-070X. DOI: 10.1016/j.tranpol.2024.06.013. URL: <https://www.sciencedirect.com/science/article/pii/S0967070X24001756> (visited on 07/01/2025).

Neubauer, J., A. Brooker, and E. Wood (July 1, 2012). "Sensitivity of Battery Electric Vehicle Economics to Drive Patterns, Vehicle Range, and Charge Strategies". In: *Journal of Power Sources* 209, pp. 269–277. ISSN: 0378-7753. DOI: 10.1016/j.jpowsour.2012.02.107. URL: <https://www.sciencedirect.com/science/article/pii/S0378775312005290> (visited on 07/03/2025).

Nylund, K. L., T. Asparouhov, and B. O. Muthén (Oct. 23, 2007). "Deciding on the Number of Classes in Latent Class Analysis and Growth Mixture Modeling: A Monte Carlo Simulation Study". In: *Structural Equation Modeling: A Multidisciplinary Journal* 14.4, pp. 535–569. ISSN: 1070-5511. DOI: 10.1080/10705510701575396. URL: <https://doi.org/10.1080/10705510701575396> (visited on 06/05/2026).

Pedrosa, G. and H. Nobre (Jan. 2018). "Second-Hand Electrical Vehicles: A First Look at the Secondary Market of Modern EVs". In: *International Journal of Electric and Hybrid Vehicles* 10.3, pp. 236–252. ISSN: 1751-4088. DOI: 10.1504/IJEHV.2018.097378. URL: <https://www.inderscienceonline.com/doi/abs/10.1504/IJEHV.2018.097378> (visited on 07/01/2025).

Recharged (2026). *Drive Electric with Confidence*. Recharged. URL: <https://recharged.com/> (visited on 06/30/2026).

Recurrent Auto (2025). *Checking Tesla Battery Health: Measure, Monitor, and Protect*. URL: <https://www.recurrentauto.com/research/check-tesla-battery-health-for-free> (visited on 06/30/2026).

Shaikh, W. A., M. A. Kalwar, M. A. Khan, A. N. Wassan, M. H. Wadho, and M. F. Shahzad (2023). "A System Dynamics Costing Model for The Refurbishment of Electric Vehicle Batteries". In: *Jordan Journal of Mechanical and Industrial Engineering* 17.01, pp. 15–32. ISSN: 1995-6665, 1995-6665. DOI: 10.59038/jjmie/170102. URL: <http://jjmie.hu.edu.jo/vol17/vol17-1/02.htm> (visited on 07/02/2025).

Sheykfhfard, A., M. Azmoodeh, S. Das, and B. Kutela (Jan. 1, 2025). "Analyzing Purchase Intentions of Used Electric Vehicles through Consumer Experiences: A Structural Equation Modeling Approach". In: *Transport Policy* 160, pp. 125–137. ISSN: 0967-070X. DOI: 10.1016/j.tranpol.2024.10.038. URL: <https://www.sciencedirect.com/science/article/pii/S0967070X24003305> (visited on 07/01/2025).

Skeete, J.-P., P. Wells, X. Dong, O. Heidrich, and G. Harper (Nov. 1, 2020). "Beyond the Event Horizon: Battery Waste, Recycling, and Sustainability in the United Kingdom Electric Vehicle Transition". In: *Energy Research & Social Science* 69, p. 101581. ISSN: 2214-6296. DOI: 10.1016/j.erss.2020.101581. URL: <https://www.sciencedirect.com/science/article/pii/S2214629620301572> (visited on 07/07/2025).

Sonar, H., H. M. Belal, C. Foropon, R. Manatkar, and V. Sonwaney (July 20, 2023). "Examining the Causal Factors of the Electric Vehicle Adoption: A Pathway to Tackle Climate Change in Resource-Constrained Environment". In: *Annals of Operations Research*. ISSN: 1572-9338. DOI: 10.1007/s10479-023-05519-8. URL: <https://doi.org/10.1007/s10479-023-05519-8> (visited on 07/01/2025).

Tankou, A., G. Bieker, and D. Hall (2023). *Scaling up Reuse and Recycling of Electric Vehicle Batteries: Assessing Challenges and Policy Approaches*.

Train, K. E. (2009). *Discrete Choice Methods with Simulation*. 2nd ed. Cambridge: Cambridge University Press. ISBN: 978-0-521-76655-5. DOI: 10.1017/CB09780511805271. URL: <https://www.cambridge.org/core/books/discrete-choice-methods-with-simulation/49CABD00F3DDDA088A8FBFAAD7E9546> (visited on 06/05/2026).

- 1 Witt, J. (2024). *Electric Car Battery Replacement Costs*. URL: https://www.recurrentauto.com/research/costs-ev-battery-replacement?utm_source=chatgpt.com (visited on 03/31/2025).
- 2
- 3 Yang, F., Y. Xie, Y. Deng, and C. Yuan (June 21, 2018). "Predictive Modeling of Battery Degradation and Greenhouse Gas Emissions from U.S.
- 4 State-Level Electric Vehicle Operation". In: *Nature Communications* 9.1, p. 2429. ISSN: 2041-1723. DOI: 10.1038/s41467-018-04826-0.
- 5 URL: <https://www.nature.com/articles/s41467-018-04826-0> (visited on 07/03/2025).
- 6 Yuan, Q., W. Hao, H. Su, G. Bing, X. Gui, and A. Safikhani (2018). "Investigation on Range Anxiety and Safety Buffer of Battery Electric Vehicle
- 7 Drivers". In: *Journal of Advanced Transportation* 2018.1, p. 8301209. ISSN: 2042-3195. DOI: 10.1155/2018/8301209. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1155/2018/8301209> (visited on 07/07/2025).
- 8
- 9 Zou, T., M. Khaleei, and D. MacKenzie (Dec. 1, 2020). "Effects of Charging Infrastructure Characteristics on Electric Vehicle Preferences of New
- 10 and Used Car Buyers in the United States". In: *Transportation Research Record* 2674.12, pp. 165–175. ISSN: 0361-1981. DOI: 10.1177/
- 11 0361198120952792. URL: <https://doi.org/10.1177/0361198120952792> (visited on 07/01/2025).